Analytics Capstone Project (ANCP)

Sparta: Wardrobe Demand Analytics for Brands

Deliverable D1: Project Charter

Group number	M
Deliverable code / version	D1 / v1
Date	13 October 2026 (due Fri 16 October 2026, 17:00 IST)
Lead owner for this deliverable	Ambarish Sengupta
Term	Term 5, 6 October to 20 November 2026
Members	C. Prashanth (BA05015) Arpan Shah (BA05009) Ambarish Sengupta (BA05007)

1. Problem statement

Business context. Sparta is a free digital wardrobe for men. Users catalogue the clothes they own, plan outfits, record purchases and keep a wishlist. With each user's opt-in consent, anonymised and aggregated data goes to brands and retailers, who today plan ranges and stock mostly on past sales and judgement. Sparta's own usage is a more natural source of wardrobe data than a questionnaire or interview, because people record what they own and want as part of using the app. This project therefore builds the app, collects data from a pilot cohort, and uses that data to build the vendor analytics.

Business decision (primary). Which menswear categories, colours and fits should a brand add or expand in its next-season range?

Decision-maker. The head of product and design at a menswear brand. The secondary decision, how much stock to commit by category, belongs to a retailer's inventory and merchandising manager.

Analytics question (one measurable sentence). Can aggregated ownership, purchase and wishlist data from at least 150 consenting Sparta pilot users, validated on a time-based holdout, rank menswear category-colour-fit segments so that the top-10 gap segments reach precision at or above 0.60 and at least 0.15 above a most-popular-segment baseline?

2. Stakeholder map

Role	Who	Interest / involvement
Decides	Head of product and design (menswear brand)	Uses the ranked gaps to set the next-season range. Primary decision-maker.
Decides (secondary)	Inventory and merchandising manager (retailer)	Uses category-level demand signals to plan stocking.
Affected	Sparta pilot users (men)	Use the app; better-fitting ranges follow; data used only with opt-in consent and only in aggregate.
Affected	Brands and retailers (vendors)	Pay for dashboards and promotions; carry the cost of wrong calls.
Supplies data	Sparta pilot users, through the app	Primary source: items, purchases, receipts, wishlist and outfits entered in the app.
Supplies data	Public datasets (Section 5)	Supporting source for seasonal history.
Guides	ANCP faculty	Feedback on each milestone; marks the deliverables.

Table 1. Stakeholder map.

Activity	Brand product head	Retailer inventory mgr	Pilot users	Project team	Faculty
Define the question and success criteria	C	C	I	A/R	C
Build and run the Sparta app pilot	I	I	C	A/R	I
Consent and contribute wardrobe data	I	I	A/R	C	-
Protect personal data (DPDP Act, 2023)	-	-	I	A/R	C
Analyse data and build the models	C	I	I	A/R	C
Decide the next-season range	A/R	I	I	C	-
Decide stock commitments by category	I	A/R	I	C	-
Review and mark deliverables	-	-	-	R	A

Table 2. RACI for stakeholders. R = responsible, A = accountable, C = consulted, I = informed, - = not involved.

3. Success metrics and acceptance criteria

Metric	Baseline compared against	Target	Acceptance test
Precision@10 of recommended gap segments	Top-10 segments by overall popularity (most owned and purchased)	0.60 or higher, and 0.15 or more above baseline	Time-based holdout: the last 10 days of app activity, unused in training or tuning; a segment is correct if it is in the holdout top quartile for new wishlist and purchase events
WAPE of segment demand forecast (seasonal pattern)	Same-period-last-year forecast	15% or more lower than baseline	Evaluated on a time-based holdout of public transaction history, which supplies the seasonal pattern
Prediction-interval coverage	Nominal 80% interval	Between 75% and 85%	Share of holdout actuals inside the interval

Metric	Baseline compared against	Target	Acceptance test
Pilot data adequacy	Planned cohort	150 or more consenting users, 6,000 or more items, 70% or more with complete taxonomy fields	Counted on the data-freeze extract of Fri 30 Oct
Reproducibility	Fresh clone of the repository	Main result reruns end to end	One command regenerates the tables and figures from versioned data

Table 3. Success metrics, baselines, targets and acceptance tests.

The project succeeds when the first four metrics meet their targets and the main result reruns. If the model does not beat the baseline, the final report will say so and state by how much.

4. Scope

In scope.

- Building the Sparta app: wardrobe catalogue (manual and photo entry), purchases with receipt upload, wishlist, outfit builder, opt-in consent flow. A Python backend with a web client first, and an API a mobile client can use.
- A rule-based outfit recommendation engine for menswear, driven by season and occasion.
- A final deployed product that pilot users reach online, and a basic aggregated view for vendors.
- Collecting pilot data in the app, then segment-level demand-gap analysis and forecasting on it.
- Time-based validation, a baseline, uncertainty estimates and a one-page decision brief.

Out of scope.

- ML-based or paid AI features, outfit inspiration matching, and the style chat assistant (planned for later).
- Womenswear and other audiences; vendor product feeds, payments and sponsored promotions.
- Reporting any individual user, or any segment with fewer than 10 contributing users during the pilot (50 in production).

Assumptions.

- Pilot users can be recruited from the team's networks and will use the app on their own, without incentives beyond the product.
- Organic app data is a valid, more natural signal of ownership and wishlist demand than self-reported surveys.
- Python, an open-source web stack and a free hosting tier are sufficient; no paid data is needed.
- Each member can give about 15 hours a week through 20 November 2026.

5. Data plan

Candidate source	Content	Access and rights	Expected size
Sparta app (primary)	Items by fixed taxonomy (category, colour, fit, material, brand, size), purchases, receipt-derived fields, wishlist, outfits	Collected from pilot users after an in-app notice and opt-in; owned by the team under that consent	150 or more users, 6,000 or more items, 1,500 or more wishlist and purchase events by 30 Oct
H&M Personalized Fashion Recommendations (Kaggle, 2022)	Transactions 2018 to 2020 and article attributes	Free with a Kaggle account; competition rules limit use to non-commercial research and education, which fits this course; terms recorded in the D2 datasheet	About 31.8 million transactions; menswear is a subset
Google Trends (menswear terms)	Weekly search interest by region	Public, free, aggregated	A few hundred weekly series
Synthetic test data	Records in the Sparta schema, for pipeline tests before pilot data arrives	Generated by the team, seeded; never used for results	About 5,000 users

Table 4. Candidate data sources.

Privacy and consent. Pilot users are adults who see a plain-language notice of what is collected and why, and opt in separately to contributing anonymised data to vendor insights; they can withdraw and delete their

data at any time (DPDP Act, 2023). The app collects only the fields it needs. Photos, receipts and accounts stay in the app's access-controlled datastore; the course repository holds only pseudonymised, aggregated extracts, with no names, contacts, images or receipts. Public data is hashed already and is not linked to app users.

6. Feasibility and risks

The project is feasible if the app is deliberately thin: only features that produce the data above, built in the first two weeks. The five largest risks follow.

#	Risk	Likelihood / impact	Mitigation
R1	The app is not live in time to gather enough pilot data before the 30 Oct freeze.	High / High	Ship the data-producing features first (catalogue, purchases, wishlist) by Wed 21 Oct; defer polish; recruit before launch; run D2 on public data plus the first pilot week.
R2	The pilot sample is small and skewed towards the team's networks.	High / Medium	Report intervals and cohort make-up; use public data for seasonality; claim segment-level patterns only, not general ones.
R3	Collecting personal data from real users breaches consent or privacy rules.	Medium / High	Notice and opt-in before any data is stored; collect minimum fields; keep identifiable data out of the repository; honour deletion requests; end-of-project retention rule.
R4	Leakage or overfitting inflates results on a small dataset.	Medium / High	Split by time; fit preprocessing inside the training window; keep the final holdout untouched until D5; compare against baselines.
R5	Scope creep and team capacity: three people build the app and the analytics together.	Medium / Medium	Freeze scope to Section 4; one lead per deliverable; weekly scope check; use the 7 to 9 November break for revision.

Table 5. Top five risks and mitigations.

Likelihood / Impact	Low	Medium	High
High		**R2**	**R1**
Medium		**R5**	**R3, R4**
Low			

Table 6. Risk matrix. Red = score 6 or 9, amber = 3 or 4, green = 1 or 2 (likelihood x impact, each scored 1 to 3).

7. Timeline

Code	Deliverable or milestone	Due (17:00 IST unless noted)	Lead owner
D1	Project charter	Fri 16 Oct	Ambarish Sengupta
-	Sparta MVP live for pilot users (project milestone)	Wed 21 Oct	C. Prashanth
D2	Data audit report, datasheet and cleaned, versioned dataset	Fri 23 Oct	C. Prashanth
D3	EDA notebook and two-page insight summary	Wed 28 Oct	Arpan Shah
-	Pilot data freeze (project milestone)	Fri 30 Oct	C. Prashanth
D4	Modelling checkpoint (baseline, candidates, experiment log)	Mon 2 Nov	C. Prashanth
D5	Evaluation memo	Thu 5 Nov	Ambarish Sengupta
D6	Deployment and monitoring plan; deployed product online	Fri 6 Nov	C. Prashanth
D7	Draft report and model card; dry-run presentation	Wed 11 Nov (written); Thu 12 Nov (dry run)	Arpan Shah
E1	Final project report	Tue 17 Nov	Ambarish Sengupta
E2	Final slide deck (PDF and PPTX)	Wed 18 Nov, 12:00	Arpan Shah
E2	Final presentation with individual viva	From 24 Nov	Arpan Shah (all present)

Table 7. Milestones D1 to E2, with the two project milestones that feed them.

8. Team roles

Each member leads three deliverables, which meets the rule that every member leads at least two. The lead owns the content, the quality check and the on-time upload; the other two members review it. Areas of contribution are a starting point, and members cover for each other as the work shifts.

Member	Roll no.	Leads	Main areas of contribution
Ambarish Sengupta	BA05007	D1, D5, E1	Product scope and priorities, user journeys, consent flow, evaluation, report and decision brief
C. Prashanth	BA05015	D2, D4, D6	System architecture, backend and data pipeline, deployment, data audit, modelling
Arpan Shah	BA05009	D3, D7, E2	Menswear taxonomy and outfit rules, pilot user recruitment and feedback, exploratory analysis, presentation

Table 8. Lead owners and areas of contribution.

Appendix A. Meeting log

Date	Attendees	Decisions	Actions
13 Oct 2026 (Meeting 1: product definition and charter planning)	C. Prashanth, Arpan Shah, Ambarish Sengupta	1. Product is a free digital wardrobe for men with an opt-in, aggregated insights layer for brands and retailers. 2. Launch scope: men only, global, Python backend, mobile and web on one backend. 3. Vendor data is opt-in, aggregated, with minimum group sizes. 4. Primary decision is product design and planning; inventory and stocking is secondary. 5. Project title: Sparta: Wardrobe Demand Analytics for Brands. 6. Building the app, its rule-based outfit engine and a deployed product is in scope; app usage is the primary source of project data.	1. Write and submit the D1 charter (Ambarish Sengupta). 2. Create the team repository with a README and draft the app architecture and data schema (C. Prashanth). 3. Define the menswear taxonomy and season and occasion outfit rules; list pilot users to recruit (Arpan Shah).

Table A1. Meeting log (date, attendees, decisions, actions).

Appendix B. Responsibility matrix

L = lead owner, R = reviewer, C = consulted on request.

Deliverable or workstream	Ambarish Sengupta	C. Prashanth	Arpan Shah
D1 Project charter	L	R	R
D2 Data audit and dataset	R	L	C
D3 EDA and insight summary	C	R	L
D4 Modelling checkpoint	R	L	C
D5 Evaluation memo	L	R	R
D6 Deployment and monitoring plan	R	L	C
D7 Draft report, model card, dry run	R	R	L
E1 Final report	L	R	R
E2 Slides and viva	R	C	L
App scope, backlog and consent flow	L	C	R
Backend, data pipeline and deployment	C	L	R
Menswear taxonomy, outfit rules, pilot recruitment	R	C	L

Table B1. Responsibility matrix.

Appendix C. AI-tool disclosure

The team used Claude (Anthropic) to structure this charter and draft its text from the team's meeting discussion and the course guide. The team checked all decisions, dates, roles and targets against the ANCP Student Guide and agreed them among the three members.

Appendix D. References

H&M Group. (2022). *H&M personalized fashion recommendations* [Data set]. Kaggle. https://www.kaggle.com/competitions/h-and-m-personalized-fashion-recommendations

Digital Personal Data Protection Act, 2023 (India).